

Recall that for $X_i \sim N(\mu, \sigma^2)$ or that if X_i is s.t. $E(X_i) = \mu$ & $V(X_i) = \sigma^2$ and n is large
 Assuming σ known. Then, by CLT $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

Recall $P(Z \leq z_{\alpha/2}) = 1 - \alpha/2$

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$$

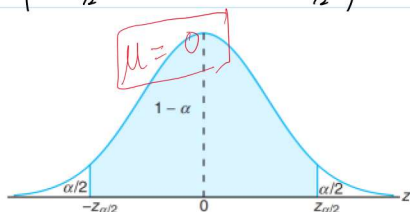


Figure 9.2: $P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$.

$$Z \sim N(0, 1)$$

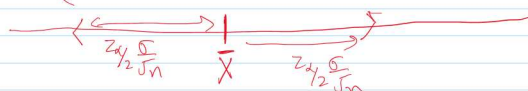
$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$$

$$P\left(-z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1 - \alpha$$

$$P\left(-z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$$P\left(-\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < -\mu < -\bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$$P\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$



$\mu \in [a, b]$ w/ prob 95% is NOT CORRECT

95% of times $[a, b]$ will cover μ ← CORRECT

Confidence Interval on μ, σ^2 Known

If $\bar{x}$ is the mean of a random sample of size n from a population with known variance σ^2 , a $100(1 - \alpha)\%$ confidence interval for μ is given by

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

where $z_{\alpha/2}$ is the z -value leaving an area of $\alpha/2$ to the right.

for $n \leq 30$ & skewed distⁿ, the approximation may not work.

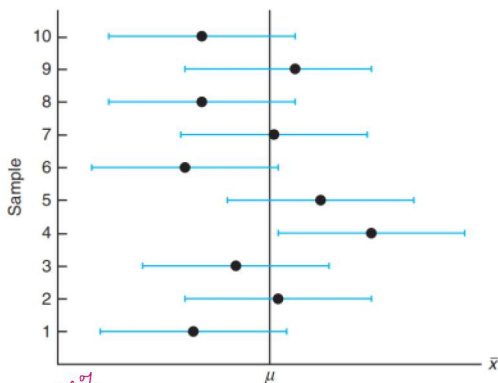
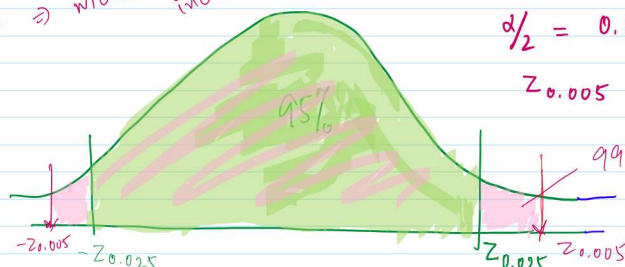


Figure 9.3: Interval estimates of μ for different samples.

$1 - \alpha = 95\%$ to 99%
 $\alpha: 5\%$ to 1%
 $z_{\alpha/2} \uparrow \Rightarrow$ width of CI increases



$$99\% \Rightarrow \alpha = 0.01$$

$\alpha/2 = 0.005$ what if we change to 99%?

$$z_{0.005}$$

$$z_{0.005}$$

$$z_{0.025}$$

$$\text{length}(99\% \text{ CI})$$

$$\text{length}(95\% \text{ CI})$$

$$\alpha = 0.05$$

Example 9.2: The average zinc concentration recovered from a sample of measurements taken in 36 different locations in a river is found to be 2.6 grams per milliliter. Find the 95% and 99% confidence intervals for the mean zinc concentration in the river. Assume that the population standard deviation is 0.3 gram per milliliter.

$$Z_{0.025} = 1.96 \quad \text{and} \quad Z_{0.005} = 2.575, \quad \bar{x} = 2.6$$

$$\sigma = 0.3$$

$$n = 36$$

$$95\% \text{ CI} \Rightarrow (2.502, 2.698)$$

$$99\% \text{ CI} \Rightarrow (2.4712, 2.73)$$

Error Margin

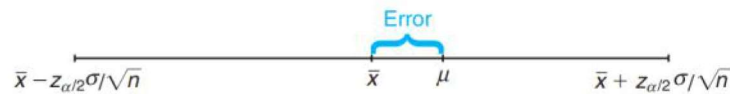


Figure 9.4: Error in estimating μ by $\bar{x}$.

Theorem 9.1: If $\bar{x}$ is used as an estimate of μ , we can be $100(1 - \alpha)\%$ confident that the error will not exceed $z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$.

SAMPLE SIZE CALCULATION

often we want to know how many samples we should collect if we want error to be less than e .

$$z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < e$$

$$\sqrt{n} > \frac{z_{\alpha/2} \sigma}{e}$$

$$n > \left(\frac{z_{\alpha/2} \sigma}{e} \right)^2$$

Theorem 9.2: If $\bar{x}$ is used as an estimate of μ , we can be $100(1 - \alpha)\%$ confident that the error will not exceed a specified amount e when the sample size is

$$n = \left(\frac{z_{\alpha/2} \sigma}{e} \right)^2.$$

Assuming σ known

Example 9.3: How large a sample is required if we want to be 95% confident that our estimate of μ in Example 9.2 is off by less than 0.05?

$$Z_{0.025} = 1.96$$

138, 2976

$$\sigma = 0.3$$

$$n = 139$$

The Case of σ Unknown

If we have $X_i \sim N(\mu, \sigma^2)$, then

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

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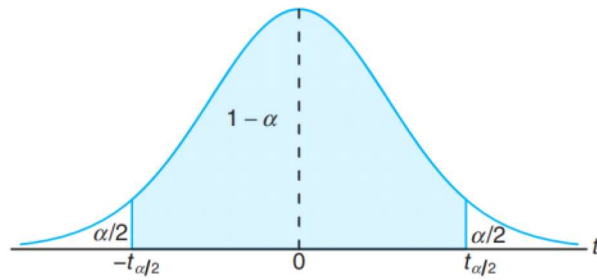


Figure 9.5: $P(-t_{\alpha/2} < T < t_{\alpha/2}) = 1 - \alpha$.

$$P(-t_{\alpha/2} < T < t_{\alpha/2}) = 1 - \alpha$$

$$\Rightarrow P\left(\bar{X} - t_{\alpha/2} \frac{s}{\sqrt{n}} < T < \bar{X} + t_{\alpha/2} \frac{s}{\sqrt{n}}\right) = 1 - \alpha$$

Confidence Interval on μ , σ^2 Unknown

If $\bar{x}$ and s are the mean and standard deviation of a random sample from a normal population with unknown variance σ^2 , a $100(1 - \alpha)\%$ confidence interval for μ is

$$\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}},$$

where $t_{\alpha/2}$ is the t -value with $v = n - 1$ degrees of freedom, leaving an area of $\alpha/2$ to the right.

Example 9.5:

The contents of seven similar containers of sulfuric acid are 9.8, 10.2, 10.4, 9.8, 10.0, 10.2, and 9.6 liters. Find a 95% confidence interval for the mean contents of all such containers, assuming an approximately normal distribution.

$$t_{0.025, 6} = 2.447$$

$$9.74 < \mu < 10.26.$$

ONE-SIDED CONFIDENCE INTERVAL

$$\rightarrow P(\mu > \mu_L) = 1 - \alpha$$

$$P(t \leq t_\alpha) = 1 - \alpha$$

$$P\left(\frac{\bar{X} - \mu}{s/\sqrt{n}} \leq t_\alpha\right) = 1 - \alpha$$

$$P\left(\bar{X} - t_\alpha \frac{s}{\sqrt{n}} \leq \mu\right) = 1 - \alpha$$



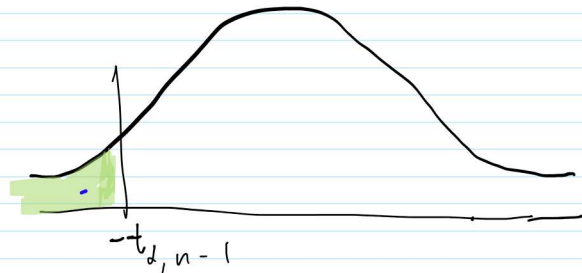
$$\mu \in \left(\bar{x} - t_{\alpha, n-1} \cdot \frac{s}{\sqrt{n}}, \infty \right) \leftarrow (1-\alpha) \% \text{ CI}$$

$$P(t > -t_{\alpha}) = 1 - \alpha$$

$$P\left(\frac{\bar{x} - \mu}{s/\sqrt{n}} > -t_{\alpha}\right) = 1 - \alpha$$

$$P\left(\bar{x} + t_{\alpha} \frac{s}{\sqrt{n}} > \mu\right) = 1 - \alpha$$

$$\mu \in \left(-\infty, \bar{x} + t_{\alpha, n-1} \frac{s}{\sqrt{n}} \right) \leftarrow (1-\alpha) \% \text{ CI}$$



LARGE - SAMPLE CONFIDENCE INTERVAL

σ unknown ; n large ($n \geq 30$) ;
population not too skewed
 s close to σ
then from CLT

$$\boxed{\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}}$$

approximate 95% CI

Example 9.6: Scholastic Aptitude Test (SAT) mathematics scores of a random sample of 500 high school seniors in the state of Texas are collected, and the sample mean and standard deviation are found to be 501 and 112, respectively. Find a 99% confidence interval on the mean SAT mathematics score for seniors in the state of Texas.

Easier to compute

$$z_{0.005} = 2.575, \quad t_{0.005, 499} = 2.585718$$

$$\bar{x} \pm z_{0.005} \frac{s}{\sqrt{n}} \Rightarrow (488.1024, 513.8976)$$

$$\bar{x} \pm t_{0.005, 499} \frac{s}{\sqrt{n}} \Rightarrow (488.0487, 513.9513)$$

STANDARD ERROR OF POINT ESTIMATOR

standard deviation (estimator) = standard error of the estimator
= s.e. (estimator)

e.g. $\bar{x} \pm z_{\alpha/2} \text{s.e.}(\bar{x})$

Confidence
Limits on μ , σ^2
Unknown

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} = \bar{x} \pm t_{\alpha/2} \text{s.e.}(\bar{x})$$

by the way, $\frac{s}{\sqrt{n}}$ really is "estimated s.e. of $\bar{x}$ "

$$\text{because } \text{s.e.}(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

But we will not care about this distinction

α, e, σ

Fix e, α

Vary σ
 n_0

$\begin{bmatrix} ? \\ \vdots \\ < \\ = \\ > \end{bmatrix}$

δ
 n_s

→ which?
& why?
compare $t_{\alpha/2, n-1}$
w/ $z_{\alpha/2}$

		sigma	50	
		error	10	
		alpha	0.05	
		z_{alpha/2}	1.96	
n		t(alpha/2, n-1)	n_t	n_z
	2	4.302653	463	96
	3	3.182446	253	
	5	2.570582	165	
	10	2.228139	124	
	20	2.085963	109	
	50	2.008559	101	
	100	1.983972	98	